APM1220_FA1

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Problem 2.2

Given the matrices:

$$A = \begin{pmatrix} -1 & 3 \\ 4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 5 \\ -4 \\ 2 \end{pmatrix}$$

(a) **5A:**

$$5A = 5 \begin{pmatrix} -1 & 3 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} -5 & 15 \\ 20 & 10 \end{pmatrix}$$

(b) **BA:**

$$BA = \begin{pmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} -1 & 3 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} -16 & 6 \\ -9 & -1 \\ 2 & -6 \end{pmatrix}$$

(c) **A'B':**

$$A'B' = \begin{pmatrix} -1 & 4 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 4 & 1 & -2 \\ -3 & -2 & 0 \end{pmatrix} = \begin{pmatrix} -16 & -9 & 2 \\ 6 & -1 & -6 \end{pmatrix}$$

(d) **C'B:**

$$C'B = \begin{pmatrix} 5 & -4 & 2 \end{pmatrix} \begin{pmatrix} 4 & -3 \\ 1 & -2 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 12 & -7 \end{pmatrix}$$

(e) **Is AB defined?** No. A is 2×2 and B is 3×2. The column dimension of A does not match the row dimension of B.

Problem 2.4

Assuming A^{-1} and B^{-1} exist, prove the following properties:

(a) **Prove that $(A')^{-1} = (A^{-1})'$:** *Proof:* By definition $AA^{-1} = I$. Taking the transpose of both sides gives $(AA^{-1})' = I'$. Using the property $(XY)' = Y'X'$, the left side expands to $(A^{-1})'A'$, and since the transpose of the identity matrix is just I, we get:

$$(A^{-1})'A' = I$$

Similarly, starting from $A^{-1}A = I$ and taking the transpose gives $A'(A^{-1})' = I$. Because both products yield the identity matrix, it follows by definition that $(A')^{-1} = (A^{-1})'$.

(b) **Prove that $(AB)^{-1} = B^{-1}A^{-1}$:** *Proof:* To show that $B^{-1}A^{-1}$ is the inverse of AB, we multiply them in both orders and verify that they simplify to I:

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

and

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I$$

Since both multiplications result in the identity matrix, the property holds: $(AB)^{-1} = B^{-1}A^{-1}$.

Problem 2.8

Given the matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$, find its eigenvalues, normalized eigenvectors, and spectral decomposition:

- **Eigenvalues:** Solve $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = \lambda^2 + \lambda - 6 = 0 \implies (\lambda+3)(\lambda-2) = 0$$

  Thus, $\lambda_1 = 2$ and $\lambda_2 = -3$.

- **Normalized Eigenvectors:**
  - For $\lambda_1 = 2$: $(A - 2I)e_1 = 0 \implies -x_1 + 2x_2 = 0$. Choosing $x_2 = 1$ gives $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, which normalizes to:

$$e_1 = \begin{pmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix}$$

  - For $\lambda_2 = -3$: $(A + 3I)e_2 = 0 \implies 2x_1 + x_2 = 0$. Choosing $x_1 = 1$ gives $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, which normalizes to:

$$e_2 = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{pmatrix}$$

- **Spectral Decomposition:** Using $A = \sum_{i=1}^{2} \lambda_i e_i e_i'$:

$$A = 2 \begin{pmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix} \begin{pmatrix} 2/\sqrt{5} & 1/\sqrt{5} \end{pmatrix} + (-3) \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{pmatrix} \begin{pmatrix} 1/\sqrt{5} & -2/\sqrt{5} \end{pmatrix}$$

$$A = 2 \begin{pmatrix} 4/5 & 2/5 \\ 2/5 & 1/5 \end{pmatrix} - 3 \begin{pmatrix} 1/5 & -2/5 \\ -2/5 & 4/5 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$$

Problem 2.11

Expanding recursively along the first row for a diagonal matrix A of size $p \times p$:

$$|A| = a_{11}A_{11} = a_{11}a_{22}\cdots a_{pp} = \prod_{i=1}^{p} a_{ii}$$

Problem 2.24

Let $\Sigma = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{pmatrix}$:

(a) $\Sigma^{-1} = \begin{pmatrix} 1/4 & 0 & 0 \\ 0 & 1/9 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

(b) Eigenvalues: $\lambda_1 = 4, \lambda_2 = 9, \lambda_3 = 1$ with standard unit eigenvectors e_1, e_2, e_3.

(c) Eigenvalues of Σ^{-1}: $1/4, 1/9, 1$ with the same eigenvectors.

Problem 2.30

Given random vector $X' = [X_1, X_2, X_3, X_4]$ with mean vector $\mu'_X = [4, 3, 2, 1]$ and covariance matrix Σ_X partitioned accordingly:

(a) $E(X^{(1)}) = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$

(b) $E(AX^{(1)}) = AE(X^{(1)}) = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = 4 + 6 = 10$

(c) $Cov(X^{(1)}) = \Sigma_{11} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$

(d) $Cov(AX^{(1)}) = A\Sigma_{11}A' = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 3 + 4 = 7$

(e) $E(X^{(2)}) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(f) $E(BX^{(2)}) = BE(X^{(2)}) = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} (1)(2) + (-2)(1) \\ (2)(2) + (-1)(1) \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$

(g) $Cov(X^{(2)}) = \Sigma_{22} = \begin{pmatrix} 9 & -2 \\ -2 & 4 \end{pmatrix}$

(h) $Cov(BX^{(2)}) = B\Sigma_{22}B' = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 9 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} 33 & 36 \\ 36 & 48 \end{pmatrix}$

(i) $Cov(X^{(1)}, X^{(2)}) = \Sigma_{12} = \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix}$

(j) $Cov(AX^{(1)}, BX^{(2)}) = A\Sigma_{12}B' = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -2 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 6 \end{pmatrix}$

Problem 2.41

Given random vector X with mean vector $\mu'_X = [3, 2, -2, 0]$ and covariance matrix $\Sigma_X = 3I_4$:

(a) **Find $E(AX)$:**

$$E(AX) = A\mu_X = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{pmatrix} \begin{pmatrix} 3 \\ 2 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \\ 3 \end{pmatrix}$$

(b) **Find $Cov(AX)$:** Using $Cov(AX) = A\Sigma_X A' = 3AA'$:

$$Cov(AX) = 3 \begin{pmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 1 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ -1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 18 & 0 \\ 0 & 0 & 36 \end{pmatrix}$$

(c) **Which pairs of linear combinations have zero covariances?** All distinct pairs of linear combinations, since all off-diagonal entries in $Cov(AX)$ are zero.